Rintaro Masaoka

Affiliation: Watanabe Group, Department of Physics, Hong Kong University of Science and Technology
Email: rmasaoka[at]ust.hk
Website: https://rintaromasaoka.github.io/

Research interests

I am interested in exploring universal properties of quantum phases of matter and quantum phase transitions. My current research focuses on frustration-free quantum systems, which exhibit distinctive properties in both gapped and gapless phases. Main research directions include:

- Proof of universal inequality on dynamical exponents for frustration-free quantum critical points.
- Formulating and characterizing frustration-free free-fermion systems.

Positions

April 2026 - Present, Research Assistant, Watanabe Group, Department of Physics, Hong Kong University of Science and Technology

Education

April 2024 - March 2026, Master of Engineering, Watanabe Group, Department of Applied Physics, University of Tokyo

April 2020 - March 2024, Bachelor of Science, Department of Physics, Faculty of Science, University of Tokyo

Awards

2025, Dean's Award, Faculty of Engineering, University of Tokyo, Department of Applied Physics, University of Tokyo

2025, Shoji Tanaka Award (Outstanding Master's Thesis in the Department of Applied Physics), Department of Applied Physics, University of Tokyo

2024, Outstanding Presentation Award (Parallel Session, Statistical mechanics), 69th Condensed Matter Physics Summer School

Grants

Research Assistantship, WINGS-FMSP Program "Forefront Physics and Mathematics Program to Drive Transformation" (FoPM), (October 2024 - February 2026)